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An Image Encryption Algorithm Based On Fractal Geometry

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Abstract

This paper introduces an effective image encryption method based on fractals for colored images. Fractals are now widely used in image security due to its randomness property and infinite boundaries. Since images are traversed over the public channel in day-to-day life confidentiality, integrity and authenticity must be pledged for prosperous transmission. This method capsulizes a larger key space and complex behavior to accomplish a productive confusion and diffusion techniques in a fruitful way. Julia set and Mandelbrot set are constructed and active confusion is exerted till the image pixels are altogether disordered and ensuing diffusion with XOR operation. Experimental results show the ability of the proposed system with respect to various statistical measures.

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Keywords: Fractals; Julia set; Mandelbrot set; Image encryption; Confusion; Diffusion.

1. Introduction

At present world visual communication plays a significant user-friendly role in information transfer such as Medical, Military, Business and so on. The information must be prevented from unauthorized access to ensure secrecy. Several techniques are available widely in securing the multimedia content in the form of audio, video and images etc. In the past few decades security has become a challenging issue for the researchers gaining its popularity. Cryptography allows encryption and decryption of images with strong mathematical background that serves hard to break the crypto-model for the intruders. Chaos based security paved a new and efficient path in image encryption due

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to its random behavior, ergodic property and sensitivity to initial conditions. Fractals also hold these properties such as pseudo-randomness, unpredictability and complex structure in a fair way. The chaos involved in the fractal geometry are efficiently utilized using the confusion and diffusion phase. The confusion phase allow the user to shuffle the pixel value position among the other pixel positions inside the image matrix to avoid correlation relation between the pixels. The diffusion phase alters the value of the original pixels leading to a completely randomized image matrix guaranteeing the optimum security.

1.1. Related work

Chaotic cryptosystem is embedded with the fractal function and chaotic map in-order to provide high amount of security to multimedia content [1, 9]. A classical cryptosystem always encloses the confusion and diffusion property to yield a fertile encryption and decryption pattern. Essaid et al. presented an improved chaotic dynamic behaviour of the Skew Tent map along with the confusion and diffusion phase [2]. A unique way of time-stamping the multimedia content using symmetric cipher system and one time key is proposed [3, 12]. Liu and Kadir designed an asymmetric cryptosystem increasing the mathematical aspect in a larger way when combined with chaos [4]. S. K. Sabnis and R. N. Awale have presented steganalysis using singular value decomposition as a steganalysis tool to measure the system [5]. Some proposed encryption model based on cascading chaotic system using various chaos maps [6, 10]. Multiple encryption schemes are proposed combining permutation-substitution technique and vector decomposition method [7, 8]. Zhang and Wei proposed a RGB image encryption scheme based on Lorenz system and DNA computation [11].

2. Fractal Geometry

Fractal is a branch of mathematics featuring self-similarity property and infinite points and it is simply denoted as a set of a recurring pattern comprising self-similarity attribute. Fractal geometry is a geometry containing irregular geometric shapes more than a combination of real and imaginary numbers. This is used in key stream generation due to the complex and random like behavior for the confusion and diffusion process.

2.1. Julia set

A Julia set called J set can be generalized and constructed in the complex plane possessing self-similarity property and infinite length. It can be mapped using the function below:

$$f(z) = (z^m + c) \quad (1)$$

here m is the real number and c is the complex number and the parameter value of c remains static in Eq. 1. The system is sensitive to the initial conditions and produces x and y coordinates in each and every iteration where x tends to be the real component and y tends to be the imaginary component. Julia set has its own importance since a minuscule change in the initial conditions can contribute to the maximum effect in the fractal generation.

2.2. Mandelbrot set

Benoit Mandelbrot, in 1979 came across a complex and perturbed structure called Mandelbrot set. The Mandelbrot set called as M set is computed with a set of points in the complex plane and mapped using the function below:

$$z_{n+1} = (z_n^2 + c) \quad (2)$$

here c and z are complex numbers in Eq. 2. Change in the parameter values can produce different fractal maps accordingly. Julia set and Mandelbrot set just rest in the thin border of chaos thus possessing the properties of dynamic behavior. The generated J set and M set are shown in Fig. 1.

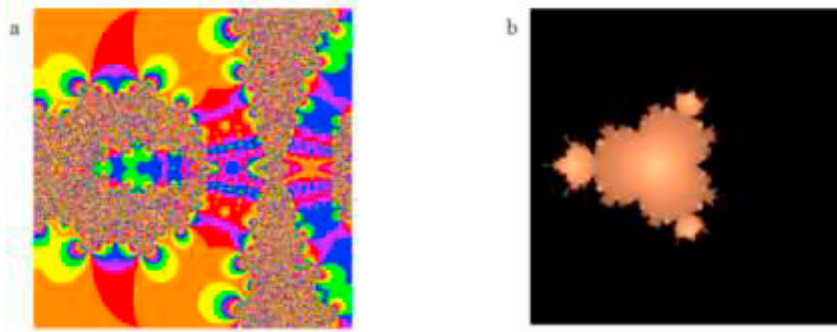


Fig. 1. (a) Julia set (b) Mandelbrot set.

3. Proposed Encryption scheme

The encryption model accomplishes two phases namely the confusion phase and diffusion phase in the encryption and decryption process. Here the plain color image 256 x 256 is considered for execution of size M x N containing M rows and N columns. The fractals Julia set and Mandelbrot set are generated for maximum number of iterations to obtain the key stream. The key stream produced by the fractals along with the keys generated using plain image is put altogether in the encryption process. The proposed encryption process flow is depicted in Fig. 2.

3.1. Encryption Algorithm

STEP 1. The original color image is separated into R-channel, G-channel and B-channel obtaining three matrices of size M x N.

STEP 2. Compute the sum of all the pixel values in the R-channel and apply modulus of 2^4 to the summed up value yielding a resultant, the same is followed for G-channel and B-channel. Three resultant values are obtained finally.

STEP 3. The resultant values got using the above step is passed as input into the Quadratic equation shown below:

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (3)$$

STEP 4. The roots or solution 's' obtained using Eq. 3. where (a,b,c) are the modulo sum of (Red, Green, Blue) channels correspondingly. s is considered as a condition parameter in the confusion phase. The Julia set is generated using Eq. 1. and the key stream j_{set} is obtained for confusion.

STEP 5. The vectors of the plain image p [$p=p_0, p_1, p_2, \dots, p_{n-1}$] and Julia set j_{set} [$j_{set}=j_0, j_1, j_2, \dots, j_{n-1}$] are compared with the conditional parameter s as a threshold value, the swap occurs only if condition is true i.e., if the threshold value is higher than the key value of the Julia set and no swap will occur if the condition is false.

STEP 6. The pixel values of the plain image are trans-located effectively using the above step yielding a confused matrix CT. Though several rounds of confusion takes place only usual amount of security can be attained. To overcome this issue diffusion is followed.

STEP 7. Now the Mandelbrot set is generated to n number of times using Eq.2. to obtain a key stream m_{set} where n is the total size of the matrix.

STEP 8. Compute the bitwise xor operation for the generated key stream m_{set} and the confused matrix CT. Additionally the second round of diffusion is performed using the Julia key stream j_{set} with the resultant matrix producing the cipher image E shown in Eq. 4. and N is the maximum intensity value of the pixel in the matrix.

$$E = \text{bitxor}(\text{bitxor}(\text{mod}(CT, N), m_{set}), j_{set}) \quad (4)$$

STEP 9. The same procedure is followed for the G-channel and B-channel. Finally the three RGB channels are fused to get the color cipher image.

3.2. Decryption algorithm

Since the proposed cryptosystem is symmetric the same set of initial values, conditional parameter are used and the decryption process is vice versa of the encryption process. The diffusion phase is first carried out and then followed by the confusion phase. Though asymmetric cryptosystem has very high robustness to the attacks symmetric crypto system has its own importance.

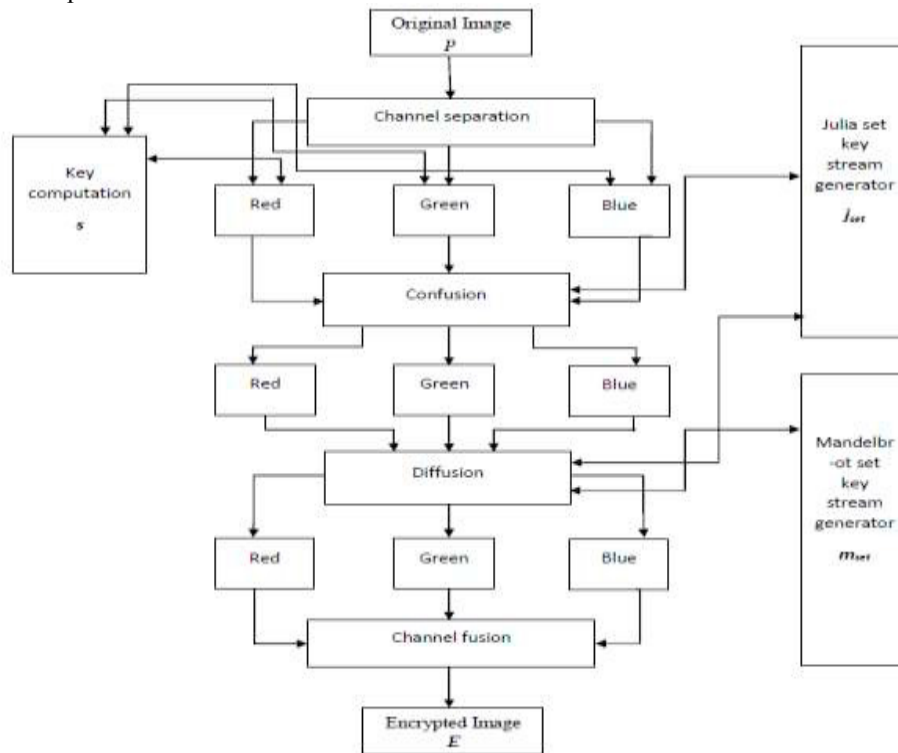


Fig. 2. Schematic flow of encryption and decryption

4. Simulation Results

To prove the practicability of the proposed method test images are obtained from the USC-SIPI Image database.

4.1. Histogram analysis

Histogram analysis is used to analyze the gray levels of the original, encrypted and decrypted images. Here 256 x 256 color image is taken to perform the analysis. To overcome the attacks the histogram of the cipher image must be uniformly distributed over the gray levels. The histogram of the R, G and B channels are individually shown with the corresponding gray levels before and after encryption. Fig. 3 shows the various histogram of the baboon image.



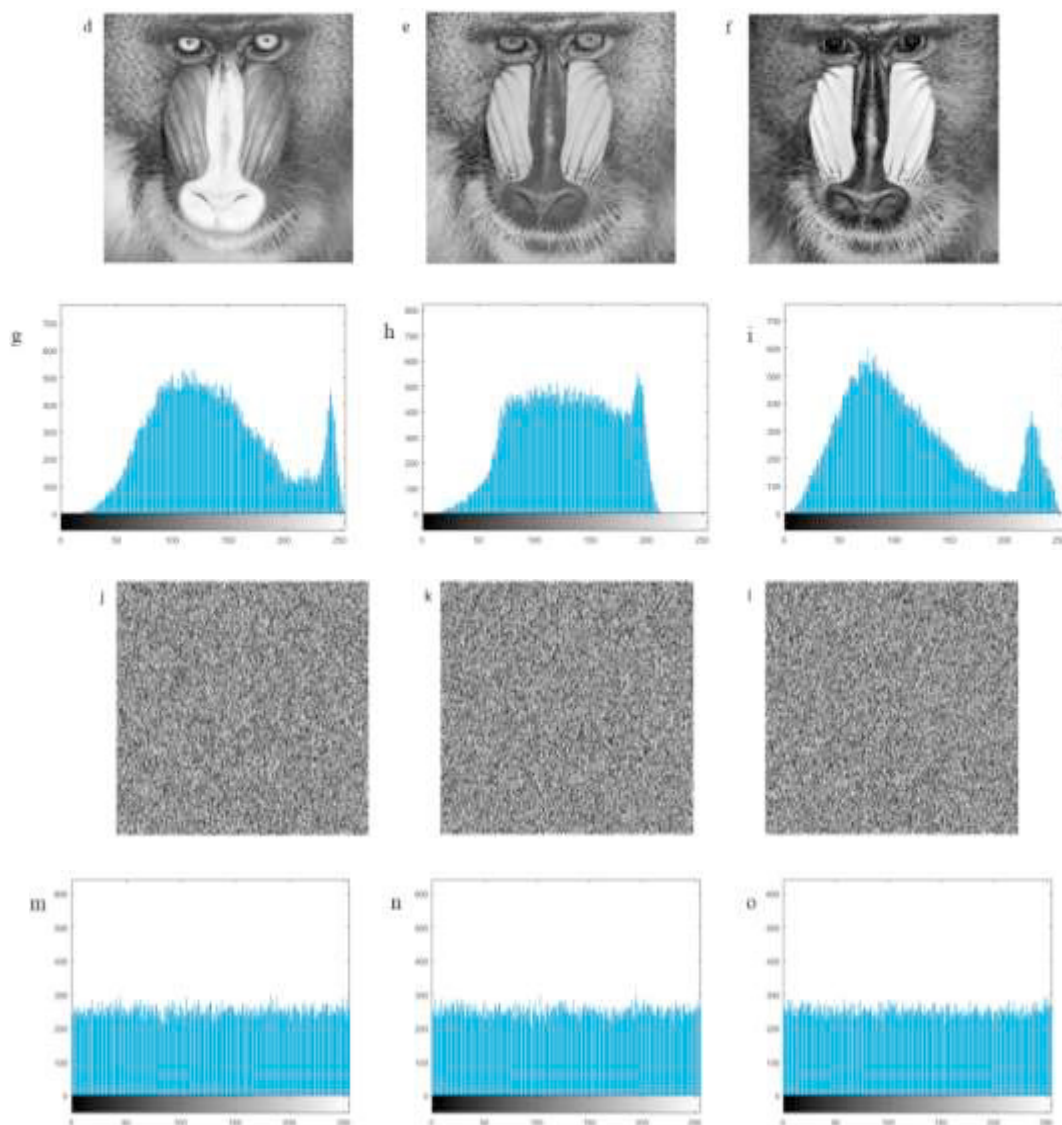


Fig. 3. Color image (a) Original, (b) Encrypted, (c) Decrypted, (d) R-channel, (e) G-channel, (f) B-channel, Histogram of channels (g) R, (h) G, (i) B, Encrypted channels (j) R, (k) G, (l) B, Histogram of the encrypted channels (m) R, (n) G, (o) B.

4.2. Information Entropy analysis

Information entropy is measured to calculate the randomness of an image. Dataset containing 256 x 256 color images are considered for simulation. Theoretically according to the research the value closer to 8 will have higher stability to attacks since the maximum gray value tends to 256 containing 2^8 symbols. An ideal random image should contain 8 as the entropy value. The entropy is calculated using the equation Eq. 5. shown below

$$H(S) = \sum_{i=0}^{2^N-1} p(s_i) \log_2 \frac{1}{p(s_i)} \quad (5)$$

where s_i denotes the probability of the symbol s . The entropy values are tabulated below in Table. 1. and the results are compared with the existing algorithms [4], [11] and [12] in Table.2. Our scheme show higher efficiency compared with the conventional schemes.

Table 1: The Information Entropy analysis

Image	Lena	Baboon	House	Peppers
Red	7.9969	7.9973	7.9968	7.9974
Green	7.9972	7.9971	7.9975	7.9972
Blue	7.9969	7.9975	7.9967	7.9973

Table 2: Comparison of Information Entropy results for Lena

Method	channel	Ref [4]	Ref [11]	Ref [12]	Proposed
Entropy	Red	7.9896	7.9968	7.9901	7.9969
	Green	7.9893	7.9965	7.9912	7.9972
	Blue	7.9896	7.9965	7.9921	7.9969

4.3. Correlation analysis

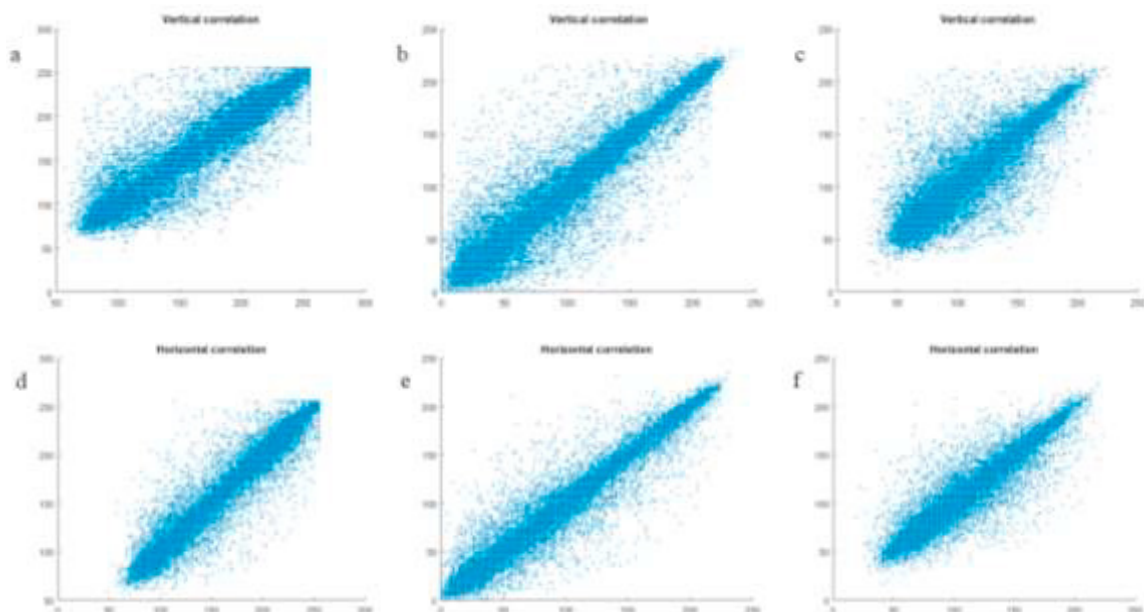
Correlation analysis is done in order to bring forth the relationship between the pixels in vertical and horizontal manner. The correlation should be close to zero to prove the encrypted result has less information about the adjacent pixel. The r component, g component and b component are separated and the correlation relationship between the adjacent pixels are measured using Eq. 6.

$$c_{xy} = \frac{\text{cov}(x, y)}{\sqrt{D(x)}\sqrt{D(y)}}$$

$$\text{cov}(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))(y_i - E(y)) \quad (6)$$

$$D(x) = \frac{1}{N} \sum_{i=1}^N (x_i - E(x))^2, \quad E(x) = \frac{1}{N} \sum_{i=1}^N (x_i)$$

The pictorial representation of the relationship between the pixels are shown in Fig. 4. demonstrating the original image and the encrypted image along with the r, g and b component. Table. 3. show the correlated results of the test image and Table. 4. show the comparison between the proposed scheme and the existing algorithm [11].



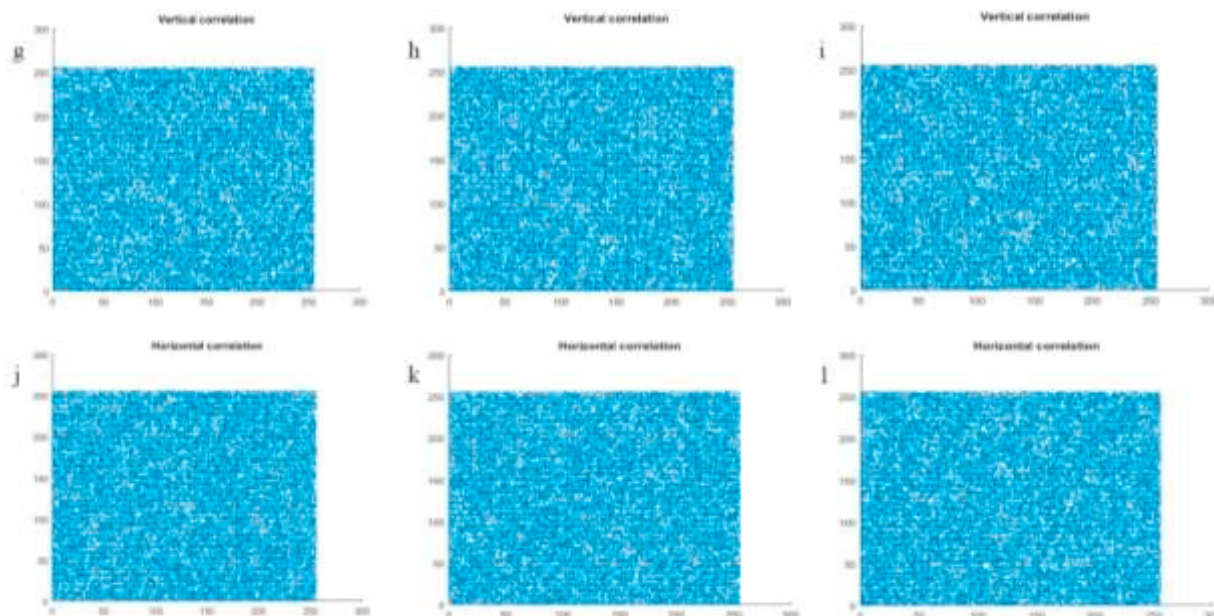


Fig. 4. Vertical relation in channels (a) R, (b) G, (c) B, Horizontal relation in channels (d) R, (e) G, (f) B, Vertical relation in the encrypted channels (g) R, (h) G, (i) B, Horizontal relation in the encrypted channels (j) R, (k) G, (l) B.

Table 3: Correlation analysis

Image	Lena		Baboon		House		Peppers	
Direction	horizontal	vertical	horizontal	vertical	horizontal	vertical	horizontal	vertical
Red	-0.0096	0.0016	0.0019	0.0084	0.0043	-0.0040	0.0076	-0.0026
Green	-0.0039	-0.0081	-0.0009	-0.0025	0.0013	-0.0049	0.0069	0.0028
Blue	0.0006	-0.0025	0.0004	-0.0048	0.0005	0.0014	0.0061	-0.0030

Table 4: Comparison of Correlation results for Lena

channel	Ref [11]		Proposed	
	horizontal	vertical	horizontal	vertical
Red	-0.0065	0.0033	-0.0096	0.0016
Green	0.0009	0.0018	-0.0039	-0.0081
Blue	-0.0008	-0.0033	0.0006	-0.0025

4.4. Key space analysis

A good image cryptosystem must possess larger key space that allows the system to be robust enough to overcome the brute-force attack. Our proposed scheme contains initial values and control parameter as keys and it is of precision 10^{10} contributing to the key space of 10^{60} which is fairly large enough for counter attacks.

4.5. PSNR analysis

Peak-Signal-Noise-Ratio analysis allows us to measure the quality of the images in decibel units (dB). Higher the ratio larger the quality and lesser ratio corresponds to low quality. Mean square error and Peak-signal-noise-ratio are balanced in such a way that increase in error results tends to lesser quality and decrease in error results tends to higher quality of the image. The psnr and mse for the encrypted images E with respect to the original images P are calculated using the formula shown below in Eq. 7 and 8. where M and N are size of the image.

$$MSE = \frac{1}{MN} \sum_{i=1}^N \sum_{j=1}^M [P(i, j) - E(i, j)]^2 \quad (7)$$

$$Psnr = 10 \times \log_{10} \frac{M \times N}{MSE} \quad (8)$$

Table 5: PSNR and MSE results

Image	Lena	Baboon	House	Peppers
PSNR	8.60	8.96	8.79	9.17
MSE	7927.52	9065.50	9325.17	7926.46

The psnr and mse results tabulated in Table. 5. ensures the image quality is high and the error is lowered describing the system to be well robust to statistical attacks.

5. Conclusion

This paper concludes introducing an image encryption algorithm based on fractal geometry generating Julia and Mandelbrot sets. An effective way of confusion and diffusion properties for color images are proposed and supported experimentally along with the simulation results such as Entropy analysis, Correlation analysis, Histogram analysis, PSNR and MSE analysis. The statistical measures exhibits positive feedback and are resistant to counter attacks.

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